The HealthInformer: RAG Model Comparison for Health & Wellness Inquiries

University of Michigan

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Educational RAG for evidence-grounded public health Q&A

1 - Problem & Goal

Why this matters

People often ask everyday health questions online, but results can be generic, misleading, or confusing. Standard LLMs can answer quickly but lack verifiable medical grounding.

HealthInformer

Our retrieval augmented generation (RAG) system pulls evidence from PubMed abstracts and rewrites it into plain language with citations. It supports entry-level wellness questions and is *not* a diagnostic tool.

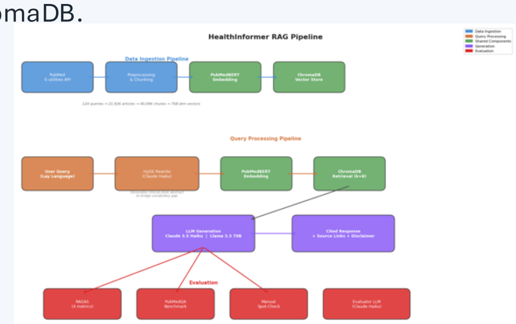
Health Questions

search engines
forums
LLMs

Need: accessible answers + evidence + citations

2 - Data & Pipeline

Corpus: 21,926 PubMed abstracts across 36 topic categories, split into 40,096 passages and indexed with BioMedBERT embeddings in ChromaDB.



3 - Evaluation & Ethics

Evaluation strategy

- RAGAS tested on 53 selected questions
- PubMedQA benchmark using 500 questions
- Manual review of 50 responses per model
- Metrics included faithfulness, answer relevancy, context precision, context recall, speed, and hallucinations

Broader impacts

- Makes peer-reviewed health information easier for everyday users to access
- Gives clear citations so users can check sources

Risks

- Users may trust answers too much
- Bias or underrepresentation in medical research
- Errors or hallucinations could still happen at scale

Safeguards

- Clear disclaimer banner
- Framed as an educational tool, not diagnosis
- No login or personal data stored

What is The Takeaway

Generation worked well once the models were given good evidence; retrieval remained the main bottleneck.

4 - Retrieval Breakthrough

Problem: everyday user questions often did not match the medical language used in research abstracts, therefore, early searches sometimes returned unrelated articles.

Solution: HyDE created a short clinical-style sample abstract before searching, which helped match questions to more relevant sources.

28% relevant pre-HyDE

66% relevant post-HyDE

5 - Main Results

Final model comparison

Metric	Claude 3.5 Haiku	Llama 3.3 70B
Faithfulness	0.898	0.880
Answer relevancy	0.914	0.898
Context precision	0.474	0.470
Context recall	0.560	0.570
PubMedQA accuracy	65.6%	77.2%
Manual quality	4.28 / 5	3.96 / 5
Hallucinations	0 / 50	0 / 50
Avg latency	13.0s	5.5s

Both models were highly grounded and showed 0 hallucinations in manual review. Llama was faster and stronger on PubMedQA, while Haiku produced cleaner, more cautious responses.

6 - Conclusion & Next Steps

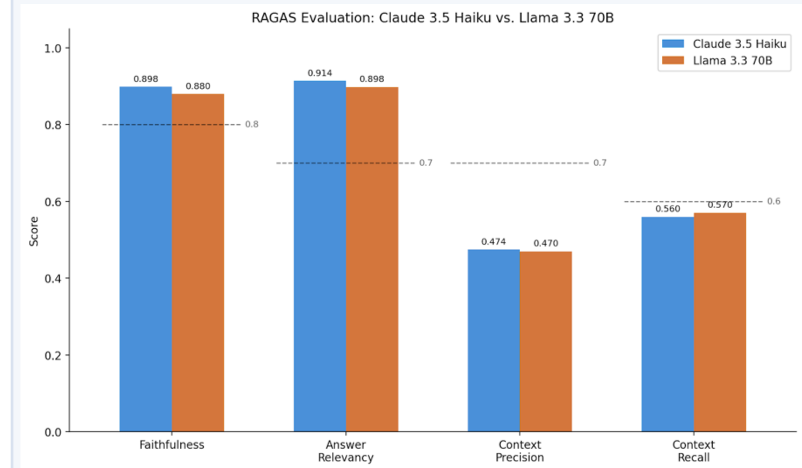
Can a RAG pipeline make peer-reviewed health evidence accessible to everyday users?

Yes, but with some important limits. Both LLM models gave reliable, cited answers when the system found the right sources. However, *future improvements* should focus on better ranking of search results, adding full-text medical articles, testing with real users, and handling questions outside the system's knowledge base.

What is The Key Insight

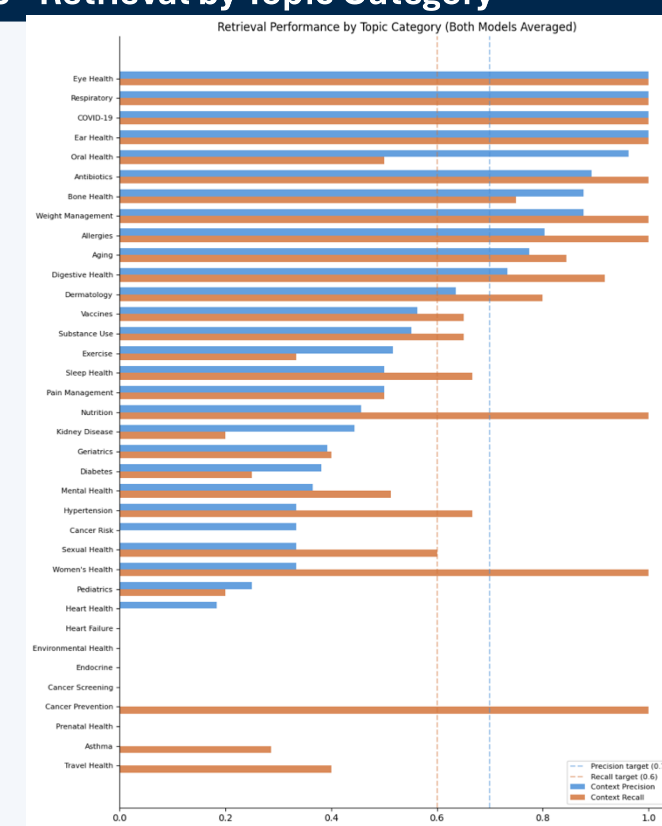
Retrieval was the limiting factor, not generation.

7 - RAGAS Performance



Both models did well at giving accurate, relevant answers, but they were weaker at finding the best supporting sources. This suggests the main issue is with retrieval, not the answer generation.

8 - Retrieval by Topic Category



Topics with clear medical terms, like vaccines and diabetes, had better search results than broader wellness topics like exercise and nutrition.